

2016.0 RANGE ROVER (LG), 303-12

INTAKE AIR DISTRIBUTION AND FILTERING - TDV6 3.0L DIESEL - GEN 2/TDV6 3.0L DIESEL

DIAGNOSIS AND TESTING

PRINCIPLES OF OPERATION

For a detailed description of the intake air distribution and filtering system and operation, refer to the relevant Description and Operation section of the workshop manual. REFER to: [Intake Air Distribution and Filtering](#) (303-12A Intake Air Distribution and Filtering - TDV6 3.0L Diesel - Gen 2/TDV6 3.0L Diesel, Description and Operation).

INSPECTION AND VERIFICATION

CAUTION:

Diagnosis by substitution from a donor vehicle is **NOT** acceptable. Substitution of control modules does not guarantee confirmation of a fault and may also cause additional faults in the vehicle being checked and/or the donor vehicle.

NOTE:

Check and rectify basic faults before beginning diagnostic routines involving pinpoint tests.

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.

Visual Inspection

MECHANICAL	ELECTRICAL
▪ Hoses and ducts condition and installation ▪ Air filter condition and installation ▪ Restricted air intake ▪ Vacuum hoses condition and installation ▪ Pipework to turbocharger condition and installation ▪ Turbocharger condition and installation ▪ Charge air coolers	▪ Fuse(s) ▪ Wiring harness ▪ Loose or corroded electrical connector(s) ▪ Mass Air Flow (MAF) sensor ▪ Air Charge Temperature (ACT) sensor ▪ Manifold absolute pressure (MAP) sensor ▪ Intake Air Temperature (IAT) sensor ▪ Intake air shut-off throttle solenoid

3. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.
4. If the cause is not visually evident, verify the symptom and refer to the Symptom Chart, alternatively check for Diagnostic Trouble Codes (DTCs) and refer to the DTC Index.

SYMPTOM CHART

SYMPTOM	POSSIBLE CAUSES	ACTION
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Vehicle does not start/hard starting	▪ Restricted/blocked air intake ▪ Restricted/blocked air filter	▪ Clear the restriction ▪ Install a new air filter as necessary
Poor performance	▪ Turbocharger fault ▪ Throttle body fault ▪ Charge air cooler hoses	▪ Inspect the turbochargers ▪ Test the intake air shutoff throttle function (make sure the throttle body returns to the open position) ▪ Inspect the charge air cooler hoses
Excessive intake noise	▪ Intake air leak after the turbocharger ▪ Intake pipe disconnected/damaged after the air cleaner ▪ Air cleaner assembly incorrectly assembled/damaged	▪ Inspect the joint between the air intake elbow and the intake air shutoff throttle. Inspect the joints between the throttle body outlets and the intake manifolds. Inspect the charge air cooler seals ▪ Check the intake system and hoses for correct installation/damage ▪ Check the air cleaner assembly for correct installation/damage

D T C I N D E X

For a list of Diagnostic Trouble Codes (DTCs) that could be logged on this vehicle, please refer to Section 100-00.

REFER to: [Diagnostic Trouble Code Index - TDV6 3.0L Diesel - Gen 1.5, DTC: Engine Control Module \(ECM\)](#) (100-00 General Information, Description and Operation).